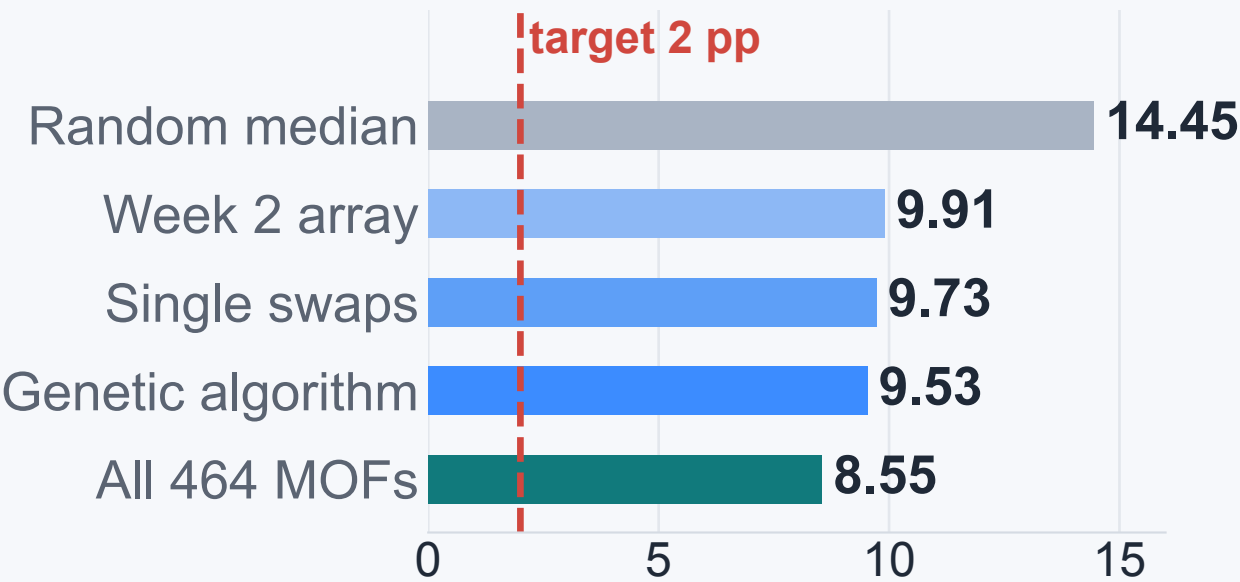


Week 3: the GA helps, but signal limits ethylene/ethane

Your question or idea	Answer
Selection method?	Greedy D-optimal design: add one of the 464 responsive MOFs at a time to maximize log det of the noise-weighted Fisher information, then swap one-for-one until nothing improves.
Detection simulation?	Signal = Henry coefficients × mole fractions, plus 1% noise. Composition is recovered by nonnegative, sum-to-one least squares on 1,000 held-out mixtures.
Genetic algorithm	104,405 of about 1.8×10^{23} possible 12-MOF arrays scored: ethylene error 9.91 to 9.53 pp, and the gain held on a fresh mixture set (10.10 to 9.52 pp).
Gradient idea	The error hinges on ethane-vs-ethylene contrast (GEHSAN's ethane coefficient is the biggest lever). Selecting with a linearized gradient did worse than the GA.

Ethylene error on held-out mixtures (pp)



Why: the limit is signal, not search

Even all 464 MOFs at once only reach 8.55 pp and still read zero ethylene in 36% of cracker-like mixtures.

Ethane outadsorbs ethylene in 87% of MOFs; the ones that favor ethylene barely adsorb. D-MOPH is physisorption-only, so metal-site ethylene binding is likely missing.

Hitting the 2 pp target would take about 0.05% sensor noise, 20x lower than assumed.

Question: what noise level is realistic for MOF sensors? Then we prune toward 4-6 MOFs.

Proposed targets: ethylene and ethane within 2 pp, ethylene/ethane ratio within 10%. Model limits: 300 K dilute-limit coefficients, assumed noise, eight-gas normalization. All designs were selected on development mixtures only.